CODENOVA AI/ML INTERNSHIP

TASK 1 REPORT

Exploratory Data Analysis of the Titanic Dataset

Data quality, passenger characteristics, and survival patterns

Submitted by

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Project Exploratory Data Analysis

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This report summarizes the executed notebook and its saved charts.

1. Abstract

This project explores the Titanic training dataset using Python and common data analysis libraries. The notebook checks the dataset structure, missing values, and duplicate rows; cleans selected missing fields; and examines survival patterns by gender and passenger class. The data contains 891 passengers and 12 columns. It has 177 missing ages, 687 missing cabin values, and 2 missing embarkation ports, with no fully duplicated rows. The observed survival rate was higher among women and first-class passengers in this dataset.

2. Objectives

• Inspect the dataset dimensions, fields, and data types.

• Measure missing values and check for duplicate records.

• Prepare a cleaned dataset with documented missing-value handling.

• Compare survival rates across gender and passenger class.

• Use charts and summary statistics to describe observed patterns.

3. Dataset and preparation

The analysis uses the Titanic training data from Kaggle. It includes passenger attributes such as age, sex, passenger class, fare, family aboard, embarkation port, and a binary survival outcome (0 = did not survive; 1 = survived). The notebook retained all 891 rows, filled missing ages with the median age, filled missing Embarked values with the most frequent port, and dropped Cabin because 687 of its values were missing. The resulting cleaned dataset has 891 rows and 11 columns, with no missing values.

Check	Observed result
Original dataset	891 rows × 12 columns
Missing Age	177 (19.9%)
Missing Cabin	687 (77.1%)
Missing Embarked	2 (0.2%)
Duplicate rows	0
After cleaning	891 rows × 11 columns; no missing values

4. Results and observations

Overall, 342 of 891 passengers survived (38.38%), while 549 (61.62%) did not. Survival was 74.20% among women and 18.89% among men. By passenger class, the observed rates were 62.96% in first class, 47.28% in second class, and 24.24% in third class. These are descriptive comparisons in the training data; they do not establish why the differences occurred.

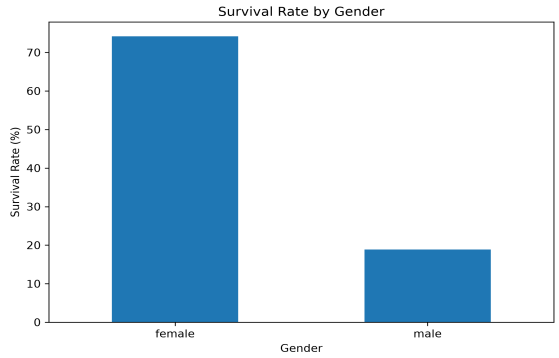


Figure 1. Observed survival rate by gender.

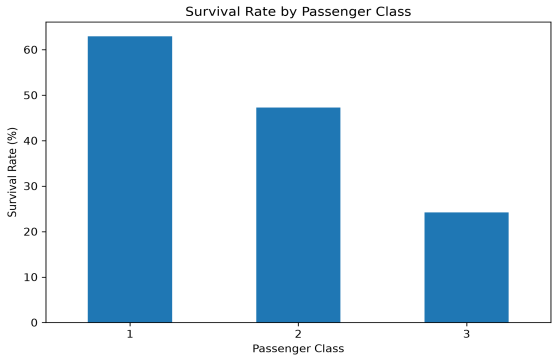


Figure 2. Observed survival rate by passenger class.

Age distribution

After filling the 177 missing ages with the median, the cleaned Age column has a median of 28 years and a mean of about 29.36 years. The plotted distribution is concentrated among younger adults, with fewer passengers at older ages. Because imputed values are included, the chart reflects the cleaned values rather than only originally observed ages.

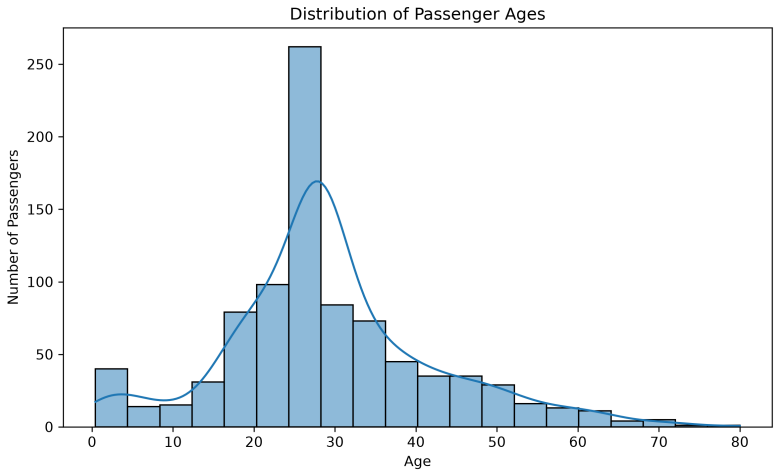


Figure 3. Age distribution after median imputation.

5. Numeric relationships

The notebook’s correlation matrix showed a correlation of about -0.34 between Pclass and Survived, and about +0.26 between Fare and Survived. These are simple linear associations in this dataset. Passenger class is encoded as an ordered number, so its correlation should be interpreted cautiously; neither correlation demonstrates causation.

6. Conclusion

The analysis identified and handled missing data, confirmed there were no fully duplicated rows, and produced a cleaned 891-row dataset. It also found clear differences in observed survival rates by gender and class. The saved figures make these group comparisons and the passenger age distribution easier to inspect. The work demonstrates practical exploratory data analysis steps that can inform later modeling.

7. Limitations

• This is an exploratory analysis of the Kaggle training set, not the full passenger population.

• Missing-data choices affect the cleaned data: Cabin was removed, and Age and Embarked values were imputed.

• Group-level differences and correlations are descriptive; they do not establish cause and effect.

• The cleaned age chart includes imputed values, which can affect its distribution.

8. Project files

The project includes the executed notebook, source data under data/, a cleaned CSV and three saved figures under outputs/, and this report. The source data is available from the Kaggle Titanic competition.

9. References

1. Kaggle, Titanic: Machine Learning from Disaster. https://www.kaggle.com/competitions/titanic/data

2. Pandas documentation. https://pandas.pydata.org/docs/

3. Matplotlib documentation. https://matplotlib.org/stable/

4. Seaborn documentation. https://seaborn.pydata.org/